

How Video See-Through MR Affects Pointing on Physical Devices: From Smartphones to Large Monitors

Yujin Jeong

Seoul National University of Science and Technology
Seoul, Republic of Korea
yujin0331@seoultech.ac.kr

HyeokJu Ko

Seoul National University of Science and Technology
Interaction Lab
Seoul, Republic of Korea
stevenko0329@seoultech.ac.kr

Jae-Yeop Jeong

Seoul National University of Science and Technology
Seoul, Republic of Korea
jaey.jeong@seoultech.ac.kr

Jin-Woo Jeong*

Department of Data Science
Seoul National University of Science and Technology
Seoul, Republic of Korea
jinw.jeong@seoultech.ac.kr

Abstract

Video See-Through (VST) mixed reality enables users to interact with physical devices while wearing head-mounted displays, but sensory barriers like optical distortion and latency may disrupt motor performance. However, the specific magnitude of this VST-induced performance cost remains unquantified. We present the first systematic investigation comparing physical device interaction between naked-eye and HMD-wearing conditions through a Fitts' Law study ($N = 21$) across five devices, from smartphones to large monitors. Results reveal non-uniform performance penalties with distinct failure mechanisms: mouse-based interactions suffered from latency-driven temporal disconnect, while touch-based interactions were hindered by optical distortion. These findings establish a quantitative baseline for VST motor performance and suggest that modality-specific compensation strategies may be necessary for effective physical device integration in mixed reality.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; *Mixed / augmented reality*; *User studies*; *Pointing*.

Keywords

Mixed Reality, Video See-Through, Pointing, Physical Device Interaction, Input Performance

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*Corresponding author



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1 Introduction

Video See-Through (VST), commonly known as pass-through, is a technology that allows users to see their physical surroundings through a headset by displaying a live, digitized version of the world captured by external cameras. This capability has transformed head-mounted displays (HMDs) into all-in-one platforms, encouraging users to integrate physical computing devices — smartphones, laptops, and monitors — into their immersive workflows [21–23]. While researchers anticipate that virtual interfaces might eventually replace physical ones [10, 21], users continue to rely on physical devices due to established digital workflows and the tactile precision of physical inputs [14]. However, integrating these tools into MR introduces a fundamental perceptual challenge. In the physical world, the performance characteristics of computing devices are well understood: larger displays enhance spatial perception [13, 25], touch interfaces provide speed advantages through direct manipulation [16, 18], and mouse-based setups deliver superior precision for detailed tasks [20]. Yet these established performance profiles may not transfer to mixed reality environments. User must transition from *direct natural vision*, which offers instantaneous feedback and infinite resolution, to a *mediated digital feed* constrained by optical distortion and system latency [3, 12]. This mediation systematically alters depth perception and hand-eye coordination [3, 29]. In this context, VST may act as an inevitable sensory barrier that disrupts the seamless motor performance users have optimized in the physical world.

Despite the qualitative awareness of these friction points, current research lacks quantitative evidence characterizing how VST viewing affects performance across different device form factors. Prior studies have identified symptomatic issues such as text unreadability or general fatigue [6, 8, 12, 24], but the specific magnitude of the **VST-induced performance cost** — the measurable penalty in speed and accuracy imposed by the medium — remains unquantified. This gap prevents researchers from determining whether the familiar performance hierarchies from physical computing (touch for speed, mouse for precision, larger displays for spatial tasks) still hold when viewed through VST feeds, or whether the optical constraints impose device-specific penalties that reshape these relationships entirely.

To address this, we present, to the best of our knowledge, the first systematic investigation quantifying the impact of VST-mediated

viewing on pointing performance across a comprehensive range of computing devices: Smartphone, Tablet, Laptop, Computer with Normal Monitor, and Computer with Large Monitor. We evaluated this efficiency through a standardized Fitts' Law pointing task. This model is a fundamental HCI metric used to characterize the speed-accuracy trade-off, predicting that the time required to reach a target is a function of its distance and width. Through a controlled study with 21 participants, we empirically measure the friction introduced by VST technology. Our findings reveal that the performance cost is not uniform across devices, identifying distinct failure mechanisms emerging for touch versus mouse inputs. These insights establish a quantitative baseline for understanding VST motor performance and reveal that physical input modalities require device-specific adaptation to function effectively in MR environments.

2 Methodology

2.1 Participants

Twenty-one students (13 males, 8 females) recruited from a local university participated in the experiment. The average age was 23.4 years ($SD = 2.26$). All participants reported having normal or corrected-to-normal vision (9 with normal vision, 12 using corrective lenses). Regarding prior VR experience, 16 participants reported having used VR devices previously, while 5 had no prior experience. Among the experienced users, 5 had used VR once or twice, and 11 had multiple experiences. Additionally, 11 participants had specific experience using pass-through devices. Participants reported a generally high interest in VR/MR technology, scoring an average of 5.15 ($SD = 1.59$) on a 7-point scale (1: Not interested; 7: Very interested).

2.2 Experimental Design

Device	Resolution	PPI	Size (in)	Rate (Hz)
Smartphone	2340 × 1080	419	6.2	60
Tablet	2800 × 1752	266	12.4	60
Laptop	2560 × 1600	178	17	60
CN	1920 × 1080	92.6	32	75
CL	3840 × 2160	102.5	43	60

Table 1: Screen specifications of devices used.

To assess the impact of VST-HMDs on performance during pointing tasks using physical devices, we employed a within-subjects design with two independent variables:

- **Device Type (5 levels):** Smartphone, Tablet, Laptop, Computer with Normal Monitor (CN), and Computer with Large Monitor (CL). Details are provided in Table 1.
- **Task Environment (2 levels):** Real World (RW) and Mixed Reality (MR).

The CL condition was included to analyze large-format displays often utilized in collaborative settings. In the MR condition, participants viewed the physical environment through a Meta Quest 3 (refresh rate; 90Hz) headset. The Meta Quest 3, a standard for consumer-grade VST, provides a baseline for current performance.

W	A	ID
50	200	2.32
50	300	2.81
25	200	3.17
30	270	3.32
25	300	3.70
15	270	4.25

Table 2: Pointing-task design.

This study establishes a foundational benchmark to measure how future hardware improvements will evolve cross-device interactions. We utilized a multidirectional tapping task based on Fitts' Law [11]. As detailed in Table 2, six Index of Difficulty (ID) levels were defined, ranging from 2.32 to 4.25 bits [9]. To ensure comparable task difficulty across devices with varying pixel densities, target sizes (W) and amplitudes (A) were standardized by adjusting pixel values to preserve the consistent physical A/W ratio across all devices. Viewing distances were controlled based on device ergonomics. Handheld devices (smartphone, tablet) allowed for natural distance adjustment [2, 30], while stationary devices were fixed at distances optimized for a 45-degree diagonal field of view (Laptop: 44cm, CN: 70cm, CL: 115cm) [1, 28]. The study was conducted under standard office lighting to reflect real-world VST performance. To control for order effects, we counterbalanced the starting environment (RW vs MR) between participants. Within each environment block, the sequence of the five devices was also counterbalanced.

2.3 Experimental Task

We adapted an open-source Fitts' law framework [27] to implement the ISO 9241-411:2012 multidirectional tapping task. Participants began by selecting a central start button, triggering nine circular targets in a radial layout. In the MR condition, participants interacted directly with the physical screens of each device while viewing them through the HMD's pass-through feed, whereas in the baseline condition, they viewed the screens directly without the HMD. Input methods were strictly controlled by form factor: direct index-finger touch for handheld devices and indirect optical mouse input for stationary devices. We assigned these modalities to match how users naturally interact with each device in daily life. This allowed us to measure the holistic performance cost of MR with ecological validity, rather than isolating hardware variables in an artificial setting.

Participants were instructed to select targets as quickly and accurately as possible. The active target appeared in red; upon selection, it turned grey, and the subsequent target was immediately highlighted. Error trials did not require correction or redoing, and the system proceeded to the next target. In total, 540 data points were collected per participant (5 Devices × 2 Environments × 6 IDs × 9 Trials).

2.4 Procedure

The experimental protocol is illustrated in Fig. 1. The study was approved by the Institutional Review Board (IRB). Upon arrival, participants were briefed on the study objectives and provided

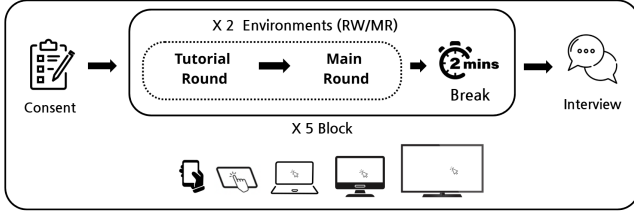


Figure 1: Experimental Protocol

written informed consent. They also completed a demographic questionnaire regarding their vision and VR experience. Prior to the MR condition, participants received instructions on fitting the Meta Quest 3 HMD and were given sufficient time to acclimate to the pass-through view to ensure visual comfort. The experiment was structured into blocks by device type. Before the main data collection for each device, a tutorial round was conducted to familiarize participants with the specific input sensitivity. Following the tutorial, participants performed the tasks under both RW and MR conditions. To mitigate physical and visual fatigue, a mandatory two-minute break was enforced between device blocks. After completing all trials, a semi-structured interview was conducted to gather qualitative feedback on visual clarity, latency perception, and overall user experience. The entire session lasted approximately 40 minutes, and participants were compensated with approximately 7.2 USD.

3 Results

We analyzed a total of 11,340 pointing and selection trials. For reliable data analysis, we screened for outliers using the three- σ rule on Movement Time (MT) and selection endpoints, but found no outliers. Subsequently, to understand our data distribution, we performed the Kolmogorov-Smirnov (KS) test. Results indicated that for the MT condition, all distributions followed a Gaussian distribution, while for ER, approximately 60% of the data followed a normal distribution. Despite the partial violation of normality assumptions in ER, we employed Repeated Measures ANOVA (RM-ANOVA) for both dependent variables, as RM-ANOVA is robust against such violations [5, 7, 26, 31]. Post-hoc pairwise comparisons were conducted using a Bonferroni-adjusted pairwise test. The results of these tests are visualized as black lines in Fig. 2 and Fig. 3.

3.1 Movement Time (MT)

As shown in Fig. 2(a) and Fig. 3a, our analysis revealed significant main effects for both **Task Environment** ($F_{1,20} = 80.61, \eta_p^2 = 0.80, p < .001$) and **Device Type** ($F_{2,42,48.48} = 147.43, \eta_p^2 = 0.88, p < .001$), as well as a significant interaction effect ($F_{3,42,68.44} = 10.57, \eta_p^2 = 0.35, p < .05$). The MR condition ($M = 733.74\text{ms}, SD = 229.69\text{ms}$) was significantly slower than the RW condition ($M = 625.96\text{ms}, SD = 184.84\text{ms}$). Notably, the performance hierarchy across devices remained consistent between environments (Fig. 3a). Comparisons indicated that touch-based mobile devices significantly outperformed mouse-based PC devices across both environments. This indicates that while VST imposes a systemic penalty, it does not fundamentally alter the relative motor difficulty inherent to each device form

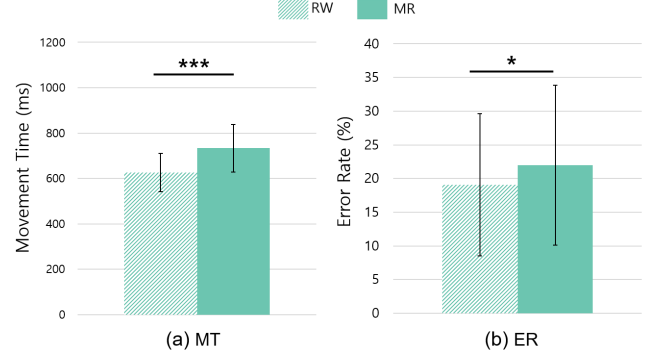


Figure 2: Overall results: (a) Mean MT. (b) Mean ER. For both measures, lower values indicate better performance. Error bars represent 95% confidence intervals. Asterisks denote statistically significant pairwise comparisons (* = $p < .05$, ** = $p < .01$, and *** = $p < .001$).

factor. However, the significant interaction effect suggests that the magnitude of this penalty was not uniform across devices. The **smartphone** was the only device resilient to VST overhead. This likely stems from its handheld nature, where direct proprioceptive cues (i.e., the physical sensation of holding and moving the device) compensated for visual latency. In contrast, the **tablet** ($p < .05$) and mouse-based devices ($p < .001$) exhibited significant degradation in MR. The performance gap was distinctly more pronounced in mouse-based interactions (**laptop**, **CN**, **CL**). This suggests that indirect input modalities (mouse) may be more vulnerable to the temporal constraints of VST than direct touch inputs. As P2 explained, this visuo-motor discrepancy hindered rapid targeting “*The touch device was much easier to control. With a mouse, however, there seemed to be a slight mismatch between my hand movements and the cursor, which made precise aiming more difficult.*” (P2). This suggests that the perceived latency of the VST pass-through may cause a disconnect between physical hand movement and digital cursor response, potentially acting as a key factor for the slower MT in PC conditions.

For PC devices, a screen size was also problematic factor: the **CL** ($M = 974.22\text{ms}, SD = 201.96\text{ms}$) was significantly slower than both the **laptop** ($M = 823.21\text{ms}, SD = 144.78\text{ms}$) and **CN** ($M = 798.55\text{ms}, SD = 156.06\text{ms}$). The significant delay in the **CL** condition, particularly in MR, was driven by additional visual search load. The large screen size required head rotation within the limited area of clear focus, which hindered rapid target acquisition. P17 highlighted how this physical constraint increased the difficulty of target acquisition “*Because the [CL] screen in MR was so large, I had to move my gaze around a lot, and sometimes I even struggled to find the mouse cursor because it wasn’t easily visible.*” (P17)

3.2 Error Rate (ER)

As shown in Fig. 2(b) and Fig. 3b, we found significant main effects for **Task Environment** ($F_{1,20} = 5.59, \eta_p^2 = 0.22, p < .05$) and **Device Type** ($F_{2,47,49.42} = 69.60, \eta_p^2 = 0.78, p < .001$), along with a significant interaction ($F_{3,42,68.43} = 3.41, \eta_p^2 = 0.15, p < .05$).

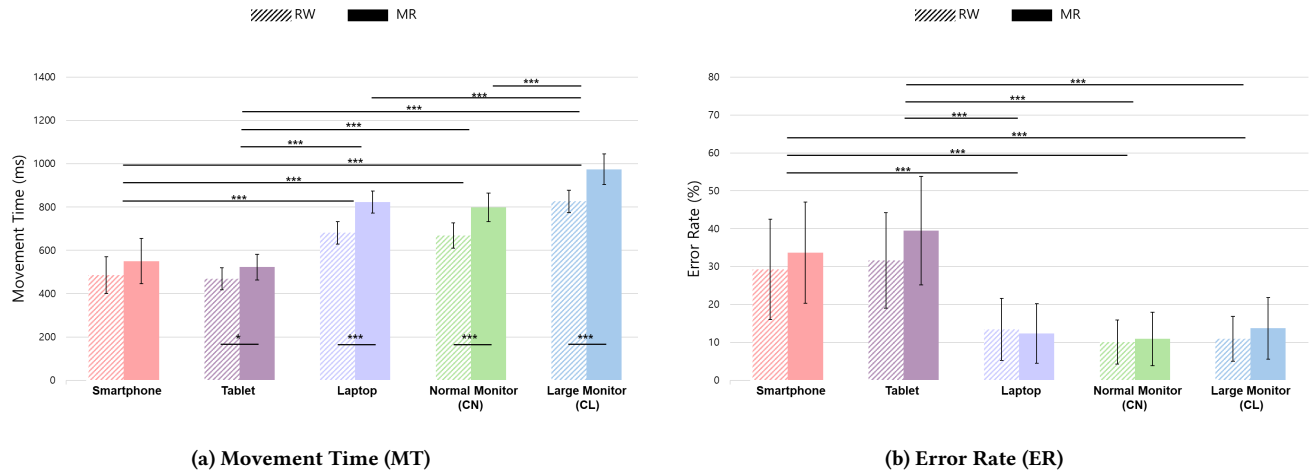


Figure 3: Device-Specific Results: (a) Mean MT. (b) Mean ER. For both measures, lower values indicate better performance. Error bars represent 95% confidence intervals. Asterisks denote statistically significant pairwise comparisons (* = $p < .05$, ** = $p < .01$, and * = $p < .001$). Comparisons between devices were computed using the combined mean values of RW and MR.**

Overall, *ER* was significantly higher in MR ($M = 22\%$, $SD = 26.1\%$) than in RW ($M = 19.1\%$, $SD = 23.2\%$).

Similar to the *MT* results, the overall performance hierarchy was consistent across environments. However, a critical distinction emerged at the device level: unlike *MT*, which significantly increased in the MR condition, *ER* remained stable regardless of the environment, showing no significant differences for any individual device. Interestingly, as shown in Fig. 3b, pairwise comparisons confirmed that there were no significant differences in error rates between RW and MR for any individual device. This suggests that the participants prioritized accuracy over speed, slowing down to mitigate visual distortions.

However, the interaction effect reveals that this accuracy degradation was not uniform. For PC devices, error rates remained low and comparable between MR and RW. This indicates a strategic speed-accuracy trade-off: participants consciously slowed down their movements in MR (as seen in the increased *MT*) to preserve accuracy. The stable coordinate system of the mouse allowed users to achieve precision by sacrificing speed. In contrast, mobile devices showed significantly higher error rates than PC devices in both environments ($p < .001$), with **tablet** ($M = 39.5\%$, $SD = 31.4\%$) and **smartphone** ($M = 33.7\%$, $SD = 29.3\%$) consistently recording the highest error rates regardless of the viewing condition. The high error rate on mobile devices is driven by the **fat-finger** problem [4, 17], likely exacerbated by VST optical distortion. Unlike PC devices, where slowing down could compensate for visual degradation, mobile touch input offers no such trade-off mechanism. The fat-finger problem imposed a physical limit on targeting precision that cannot be overcome through strategic timing adjustments alone. Participants reported that these distortions were not static but dynamic ($N=18$); P5 noted, *"The screen size also fluctuated unpredictably, making it feel unstable"* which made precise tapping on small targets (15-25px) exceptionally difficult. Screen size amplified this effect. P7 highlighted that *"the distortion... was even stronger on*

smaller screens" ($N=9$), confirming that the VST rendering struggles with the high-frequency details of compact mobile displays. Consequently, the ergonomic advantage of touch input diminished for precision tasks. As P17 explained, *"Once the targets got smaller than my fingertip, it didn't feel any easier than using a mouse"* ($N=4$), suggesting that when VST visual noise masks the target, the directness of touch input becomes a liability.

4 Discussion and Conclusion

In this study, we investigated VST performance overhead across diverse computing devices and found several critical insights. First, the performance differences among devices remained consistent across environments, indicating that the fundamental input characteristics of each device were maintained in MR. Second, despite this consistent trend, overall performance in the MR was substantially worse than in the RW environment. VST introduced distinct challenges depending on the input modality. The observed trends suggest that performance costs may take the form of a **temporal disconnect** for mouse users potentially linked to system latency, whereas touch users may encounter **spatial instability** likely attributed to optical distortion [3]. While future research utilizing objective hardware metrics is required to confirm these causal mechanisms, these patterns imply that a one-size-fits-all correction/adaptation algorithm may be insufficient. From a design perspective, this points toward the need for modality-specific compensation, such as predictive cursor trajectories for indirect input or localized distortion correction for direct touch zones.

Furthermore, the poor performance of the CL condition highlights a critical scalability issue in VST. Unlike physical environments where large screens typically improve targeting [13], our result shows that the extensive head rotations required to scan large displays likely expose users to increased motion-to-photon latency [15, 19]. To address this, MR systems could implement adaptive mirroring triggered by postural thresholds. When the system

detects head rotations or gaze angles exceeding a certain ergonomic threshold, it could dynamically generate a virtual, stabilized mirror of the physical content within the user's central field of vision.

Notably, a majority of our participants (16/21) were experienced VR users, with 11 specifically having prior experience with pass-through devices. The fact that substantial physical and visual friction emerged, while they are relatively experienced users, suggests that these constraints are not merely issues of unfamiliarity but reflect inherent limitations of current VST hardware and rendering pipelines. Therefore, for novices who lack prior adaptation strategies and typically have lower initial tolerance for sensory conflicts, the physical strain and perceptual friction reported here would likely be even more severe.

Finally, while the intrinsic characteristics of each device remain consistent across environments, VST-induced constraints profoundly impacted user experience even in simple Fitts' Law tasks. This suggests that more complex, attention-demanding everyday activities may suffer from compounded disruptions due to high cognitive load and prolonged visual fatigue. Future research should investigate these effects in multifaceted, high-load scenarios to determine if users can eventually habituate to VST artifacts or if performance degradation remains a persistent barrier to long-term adoption. In addition, future work may consider expanding the error-rate analysis by categorizing specific error types to better understand VST's impact on precision. Alternative statistical approaches, such as linear mixed-effects models, could also allow for a more nuanced understanding of individual variability and the interactions between device modalities and VST artifacts.

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